



The Flowering Boundary

A Developmental Phase Transition Extended Through Living Matter



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The switch from leaf growth to flowering is an unusually visible case of a living system changing its collective organization while remaining materially continuous. The Flowering Boundary treats the transition interval itself as a measurable dynamical object, with duration, spatial extent, recovery behavior, and history dependence. Because the shoot tip keeps growing while its developmental state changes, growth can preserve a spatial record of a brief developmental transition.

Measuring the switch to flowering

What does the transition to flowering look like when the transition interval itself is measured as a dynamical object?

How the switch works

1. **A multistable network supports distinct regimes.** At the shoot apex, environmental and endogenous signals act through a multistable gene-regulatory network whose dynamics support distinct developmental regimes.
2. **The transition interval is an object.** The transition interval itself is a measurable dynamical object with duration, spatial extent, recovery behavior, and history dependence.
3. **A coordinate for progression.** A coarse developmental coordinate, Φ , represents progression through the changing regime. Perturbation and recovery measurements determine which organization the system tends to restore at different stages of commitment.
4. **Temporal progression becomes spatial structure.** Because the meristem continues to grow while its developmental state changes, temporal progression can be converted into spatial structure.
5. **State, stability, and retained history are three different properties.** Present developmental state is distinct from the restorative stability of that state and from the predictive contribution of physically retained developmental history.
6. **Retained history is physical.** Earlier events can alter transcriptional state, chromatin configuration, hormonal sensitivity, geometry, cell identity, and regulatory relationships, so similar present conditions can produce different subsequent trajectories.
7. **Transitions become structure.** Because organization changes while new material is being produced, the consequences of a transition can become incorporated into the structures that follow. Living development builds its transitions into lasting structure.

Terms to know

- **Shoot apex and meristem.** The growing tip of a plant shoot and its tissue of dividing cells, where leaves or flowers are produced.
- **Multistable network.** A gene-regulatory network with more than one stable regime it can settle into.
- **Φ .** A coarse developmental coordinate that represents progression through the changing regime. In plain terms: how far along the switch to flowering the tissue

has moved.

- **R.** Local restorative dominance under a specified perturbation class. In plain terms: after a nudge of a given kind, how strongly the tissue returns to the organization it was holding.
- **CHpred.** The additional predictive contribution of retained history. In plain terms: how much better future development is predicted when history-bearing variables are added to everything else measured now.
- **Observational grain.** The level of detail at which the system is measured. CHpred is defined at a specified grain.

Three things to measure

- Φ describes the collective developmental regime.
- **R** describes local restorative dominance under a specified perturbation class.
- **CHpred** describes the additional predictive contribution of retained history.

Retained history is operationalized at a specified observational grain by asking whether designated history-bearing variables improve prediction of future development beyond other presently measured state variables and environmental inputs.

What experiments can check

- **The central prediction.** Transition duration and growth rate systematically influence the anatomical extent over which intermediate or mixed developmental organization appears.
- **Commitment, measured.** The behavior of Φ , **R**, and **CHpred** across the transition provides a testable description of developmental commitment, open to confirmation or revision through experiment.

What flowering reveals

Flowering is a concrete model of how a continuous living system can change what it preserves, how it responds to perturbation, and how its own past remains physically consequential in the construction of its future.

The bigger picture

The same continuity appears in a growing plant. A developmental transition becomes part of the structure that follows, so the plant’s own past stays physically consequential in what it builds next.

Full paper on PhilArchive: [The Flowering Boundary: A Developmental Phase Transition Extended Through Living Matter](#)

Before this: [New Bodies, Old Capacities](#).

Next: [The Period Lattice](#), the same relation in a small exact model.

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